

b-secure Guard Mobile App

The **b-secure Guard App** is the professional-facing mobile application for verified security guard personnel on the b-secure platform. Guards use this app to manage their availability, receive and fulfill booking requests from users, navigate to job sites, track real-time earnings, maintain compliance documents, and respond to SOS emergencies.

Purpose

Security guard professionals use this app to:

- Toggle their **online/offline availability** to receive booking requests
 - Receive **real-time push notifications and WebSocket alerts** for incoming requests
 - **Accept or decline** bookings within a 30-second window
 - **Navigate** to the user's location using native maps integration
 - Progress through booking **status steps**: En Route → Arrived → Started → Completed
 - Monitor **earnings**, request payouts, and view payout history
 - Upload and manage **compliance documents** (Aadhaar, PAN, police verification, etc.)
 - Complete an **onboarding flow** that gates entry to the main app until admin-approved
 - Trigger and receive **SOS alerts** with full-screen priority notifications
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Tech Stack

Layer	Technology
Framework	React Native via Expo SDK 51
Language	TypeScript (strict mode)
Navigation	Expo Router v3 (file-based routing)
Server State	TanStack Query v5 (@tanstack/react-query)

Layer	Technology
Client State	Zustand v4
Maps	React Native Maps (react-native-maps)
Real-time	WebSocket (native browser API)
Push Notifications	Expo Notifications (expo-notifications)
Background Location	Expo Location + Expo Task Manager (expo-task-manager)
Styling	NativeWind v4 (Tailwind CSS for RN)
Forms	React Hook Form + Zod
HTTP Client	Axios
Media	expo-image-picker , expo-av
Storage	expo-secure-store (tokens), AsyncStorage

Folder Structure

```

guard-app/
├── app/
│   └── routes
│       ├── _layout.tsx
│       └── providers
│           ├── index.tsx
│           └── login or tabs
│               ├── (auth)/
│               │   ├── _layout.tsx
│               │   ├── login.tsx
│               │   └── otp.tsx
│               └── onboarding/
│                   ├── _layout.tsx
│                   └── personal-info.tsx
│   └── skills
│       ├── documents.tsx
│       └── complete.tsx
│   └── screen
│       ├── (tabs)/
│       │   ├── _layout.tsx
│       │   └── dashboard.tsx
│       └── radar
│           └── earnings.tsx
│   └── payout
│       ├── history.tsx
│       └── profile.tsx

```

Expo Router file-based

Root layout – auth gate,

Redirect: auth check →

Phone number entry

OTP verification

Step 1: name, DOB, city,

Step 2: document uploads

Step 3: under-review

Bottom tab navigator

Online toggle + booking

Earnings summary +

Booking history

Profile + documents +

```

schedule
|   | booking/
|   | | request.tsx
|   | | # Full-screen incoming
request modal
|   | | active.tsx
|   | | # Active booking map +
steps
|   | | [id].tsx
|   | | # Completed booking detail
|   | | documents/
|   | | | upload.tsx
|   | | # Document camera/upload
flow
|   | src/
|   | | api/
|   | | # Axios instance + service
modules
|   | | client.ts
|   | | guardProfileService.ts
|   | | bookingService.ts
|   | | earningsService.ts
|   | | documentService.ts
|   | | locationService.ts
|   | | onboardingService.ts
|   | | store/
|   | | | guardStore.ts
|   | | | activeBookingStore.ts
|   | | | earningsStore.ts
|   | | | # Zustand stores
|   | | | hooks/
|   | | | | useGuardStatus.ts
|   | | | | useIncomingRequest.ts
|   | | | | useAcceptBooking.ts
|   | | | | useActiveBooking.ts
|   | | | | useNavigationToUser.ts
|   | | | | useEarnings.ts
|   | | | | useDocumentUpload.ts
|   | | | | useBookingTimer.ts
|   | | | # Custom React hooks
|   | | | services/
|   | | | | websocketService.ts
|   | | | | backgroundLocation.ts
|   | | | | # Guard WebSocket client
|   | | | | # BG location module
|   | | | types/
|   | | | | guard.ts
|   | | | | # All TypeScript
interfaces
|   | | constants/
|   | | | tasks.ts
|   | | | # Task name constants
|   | | | config.ts
|   | | | # API URL, WS URL,
thresholds
|   | | components/
|   | | | # Shared UI components
|   | | | RadarAnimation.tsx
|   | | | BookingRequestCard.tsx
|   | | | StepProgressBar.tsx
|   | | | EarningsChart.tsx
|   | | | DocumentItem.tsx
|   | | | SOSButton.tsx
|   | | assets/
|   | | | sounds/
|   | | | | incoming_request.mp3
|   | | | images/

```

- app.json
- babel.config.js
- tailwind.config.js
- tsconfig.json
- package.json

Documentation Index

File	Description
<u>overview.md</u>	Setup, env vars, background location config, auth gate, Zustand stores
<u>screens.md</u>	All screen components with full TSX code
<u>background_location.md</u>	Background location tracking deep-dive
<u>hooks.md</u>	All custom hooks with TypeScript code
<u>realtime.md</u>	WebSocket service, message types, notification handling
<u>api_integration.md</u>	Axios setup, all service modules, TypeScript interfaces

Quick Start

Clone and install

```
git clone https://github.com/bsecure/guard-app
cd guard-app
npm install
```

Copy env

```
cp .env.example .env.local
# Fill in API_URL, WS_URL, etc.
```

Start

```
npx expo start
```

Physical device build (required for background location)

```
npx expo run:ios --device
npx expo run:android --device
```

Important: Background location tracking requires a physical device. Always test location features using a custom development build, not Expo Go.

Key Differentiators from User App

Feature	Guard App	User App
Background Location	Broadcasts every 10s via expo-task-manager	Not required
Online/Offline Toggle	Explicit availability control	Always discoverable
Verification Gate	Onboarding + admin approval required	Instant registration
SOS Receiver	Receives SOS from users, full-screen alert	Sends SOS
Earnings & Payouts	Full earnings dashboard + payout requests	Not applicable